Optical Transport Failure Resolution for Huawei Access Node JAY-ACC-60 at Jayanagar Metro Site

Node: JAY-ACC-60 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6016

Description

On the Huawei Access Node JAY-ACC-60 at Jayanagar Metro Site, an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) occurs due to the degradation of signal strength in the optical transport domain. This is typically caused by the degradation of the SFP (Small Form-factor Pluggable) or XFP (10G Small Form-factor Pluggable) transceivers, which are responsible for transmitting and receiving optical signals over the fiber optic cables. The failure of the optical transport domain can be attributed to various factors such as fiber cuts, SFP/XFP degradation, or Rx/Tx power loss. This failure can have a cascading effect on the network, leading to increased packet loss, jitter, and latency. It is essential to identify and rectify the root cause of the failure to prevent further network degradation. The Huawei Access Node JAY-ACC-60 uses the NE2000 chipset, which is responsible for the optical transport domain. The NE2000 chipset uses the ITU-T G.709 standard for Optical Transport Network (OTN) protocol. The failure of the optical transport domain can be diagnosed using the Huawei CLI, which provides a comprehensive set of commands to troubleshoot and resolve the issue.

Pre-requisite Checks

- Verify that the site is accessible and safe for maintenance.
- Check the redundant link status to ensure that the network is not affected by the failure.
- Verify that the correct login credentials are used to access the Huawei CLI.
- Backup the configuration before making any changes.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display transceiver info, Expected Output: Transceiver Info: Module 1: SFP-1000BASE-LX10 Module 2: XFP-10G-LR , Description: This command displays the transceiver information, including the type and status of the SFP or XFP modules.

Command: display optical-performance monitor, Expected Output: Optical Performance Monitor: Module 1: Rx Power: -30.5 dBm, Tx Power: -10.5 dBm Module 2: Rx Power: -35.5 dBm, Tx Power: -15.5 dBm , Description:
```

```
This command displays the optical performance monitor, including the
Rx and Tx power levels of the SFP or XFP modules.
```

Resolution Steps

Step 1: Step: Replace the SFP or XFP module, Command: replace transceiver module 1, Description: Replace the SFP or XFP module with a new one to resolve the issue.

Step 2: Step: Check the fiber optic cables, Command: display fiber optic cable status, Description: Check the status of the fiber optic cables to ensure that they are not damaged or cut.

Post-Check Verification

```
Command: display transceiver info, Expected Output: Transceiver Info:
Module 1: SFP-1000BASE-LX10 Module 2: XFP-10G-LR , Description: Verify
that the transceiver information is displayed correctly.

Command: display optical-performance monitor, Expected Output: Optical
Performance Monitor: Module 1: Rx Power: -30.5 dBm, Tx Power: -10.5
dBm Module 2: Rx Power: -35.5 dBm, Tx Power: -15.5 dBm , Description:
Verify that the optical performance monitor is displayed correctly.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: restore configuration, Description: Restore the original configuration to revert any changes made during the resolution process.

- Step: Replace the replaced SFP or XFP module, Command: replace transceiver module 1, Description: Replace the replaced SFP or XFP module with the original one to revert the changes made during the resolution process.